

Less to Lose: How Shorter and More Flexible Degrees Reshaped University Choice

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Abstract

This paper examines how students respond when university degrees become shorter, exploiting Portugal's Bologna Process implementation as a natural experiment. Using a regression discontinuity design that compares cohorts just before and after the reform, I document a 9% increase in university attainment and a 0.14 increase in years of schooling on net. Individuals reallocate across fields, from healthcare programs (30% decline) toward STEM (22% increase) and economics (10% increase), with stronger effects for females. Post-reform cohorts forgo earnings at younger ages due to extended time in education, but overtake pre-reform cohorts by their early thirties as human capital gains more than offset the initial losses. This is driven by an increase in hourly wages, which I estimate to be 1.6%. Using an AKM decomposition, I find that 30% of the wage effects operate solely through the workplace channel. Overall, gains from expanding university access and reallocating students toward higher-paying fields outweigh losses from degree compression, validating the less is more hypothesis.

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1 Introduction

How do students respond when the time required to complete university degrees changes? While an extensive literature examines the effects of extending compulsory schooling—typically through reforms that raise the minimum school-leaving age ([Angrist and Krueger, 1991](#); [Meghir and Palme, 1999](#); [Acemoglu and Angrist, 2000](#); [Oreopoulos, 2007](#); [Pischke and Von Wachter, 2008](#); [Portugal et al., 2024](#))—we know surprisingly little about what happens when higher education degrees become shorter. This gap matters for education policy worldwide, as countries have extensively debated whether the gains from shorter degrees attracting more students to university outweigh the potential human capital losses from compressing the same curricular content into fewer years of study. The fundamental question is whether shorter degrees achieve the “less is more” outcome, where reducing time requirements actually increases total human capital by attracting students who would otherwise not attend university. Understanding how students navigate this trade-off when degree lengths change and what the consequences are for their careers is crucial, yet empirical evidence remains scarce.

The Bologna Process, which required over 20 countries to compress their university degrees, provides unprecedented variation to answer this question. This reform remains largely unstudied ([Kroher et al., 2021](#)), despite being Europe’s most comprehensive higher education restructuring. Countries transitioned from Continental European models requiring 4-5 years for a first degree to the Anglo-Saxon model offering a 3-year bachelor’s degree followed by an optional 2-year master’s. This comprehensive restructuring across an entire continent offers three key advantages for identification: the reform’s near-universal adoption within countries eliminated selection into treatment, the rapid implementation timeline created clean discontinuities in exposure, and the magnitude of the change—reducing minimum degree requirements by 25-40%—ensures sufficient variation to detect behavioral responses.

The net effect of shorter degrees is theoretically ambiguous because the reform simultaneously shifts two opposing margins, with implications for (i) educational attainment, (ii) field choices, (iii) earnings, and (iv) career trajectories. On the extensive margin, shorter degrees lower barriers to university entry: students who previously found a 4- to 5-year commitment too costly might now enroll in the more accessible 3-year program. While these marginal students gain access to higher-paying jobs requiring university credentials,

they must delay labor-market entry and forego earnings during their studies. On the intensive margin, the reform affects students who would have attended university regardless in multiple ways. On the one hand, it creates an earlier exit option: students who would have completed 4-5 years might now stop after 3 years, entering the labor market sooner and accumulating more work experience by any given age, but potentially sacrificing human capital from the foregone years of education. On the other hand, students can now get a master’s degree in the same time as a pre-reform bachelor’s degree, and enter the labour market at the same age but with a more valuable degree. At the same time, the reform can cause field reallocation: students might shift into fields that were previously too time-intensive, reallocating across disciplines and potentially accessing higher wage growth if moving to higher-return fields. Which effect dominates—and thus whether the “less is more” hypothesis holds—is ultimately an empirical question that speaks directly to the optimal design of higher education systems.

Portugal offers exceptional conditions to identify these effects. The reform implementation was remarkably sharp, with almost all programs restructured between 2006/07 and 2007/08. Portugal’s standard university entry age of 18 implies that cohorts turning 18 in 2007 or later faced the new system when making enrollment decisions, while older cohorts had already decided under the previous system. Importantly, I find no evidence of strategic delay into university around the reform. Combined with Portugal’s comprehensive linked employer-employee data (Quadros de Pessoal) that tracks educational attainment and labor market outcomes, I can trace how the reform affected both educational outcomes and career trajectories. To address potential selection into private sector employment, I complement the analysis with Labor Force Survey (LFS) data covering the entire population.

I find that the “less is more” hypothesis holds. On the education side, the Bologna reform increased university graduation rates by 2.1 percentage points ($\sim 8\text{-}9\%$ relative to baseline), driven primarily by an uptake of shorter degrees. The extensive margin dominates: proxies for average years of schooling indicate an increase of 0.14 in the population. The reform also induced substantial field reallocation. Students shifted away from healthcare¹ programs ($\sim 30\%$ decline)—where, uniquely, Bologna actually lengthened degrees to meet European standards—toward STEM ($\sim 22\%$ increase) and Economics ($\sim 10\%$ increase) that became

¹Excluding medicine

more accessible with shorter requirements. These effects are particularly pronounced for females, who show both stronger field reallocation and higher overall graduation increases. On the labor market side, I find that cumulative earnings are initially lower for post-reform cohorts but higher by the early thirties. This pattern is consistent with an education-investment tradeoff: by remaining in school longer, individuals forgo earnings at younger ages, but the losses are more than offset by subsequent human capital gains. In line with this, I find that hourly wages increase by 1.6%; a back-of-the envelope calculation merging wage and schooling effects implies a 11.4% return per year of university education. Using an AKM decomposition following [Portugal et al. \(2024\)](#), I find that 30% of wage gains operate through firm sorting, suggesting the reform enhanced graduates' skill sets despite degree compression but also allowed them to reach better paying firms. Women experience steeper early-career wage growth in the first three years of their career, with similar effects among those who change firm and those who remain in the same firm.

Finally, I explore two complementary approaches providing suggestive evidence to disentangle effects by socio-economic status. The first method uses early labour market entry (before age 18) as a proxy for low-SES individuals. The second constructs Lee bounds on wage effects, trimming the post-reform wage distribution under alternative assumptions about whether the marginal students brought into university by the reform come from the top or bottom of the earnings distribution. Among low-SES individuals, university completion rises by more than 60% for females, driven mostly by economics degrees, and hourly wages increase by 3.6%. The concentration in economics is consistent with these students responding to the new accessibility of a shorter degree with strong labour market returns. Lee bounds highlight that while wages are initially lower in the mid-twenties, the penalty fades by around age 30.

This paper makes three main contributions. First, I provide the first evidence on how the Bologna Process affected labour market outcomes — graduation rates, field composition, earnings, and career trajectories — moving beyond existing work confined to enrollment effects ([Argentin and Triventi, 2011](#); [Brunori et al., 2013](#); [Cappellari and Lucifora, 2009](#); [Di Pietro and Cutillo, 2008](#); [Bondonio and Berton, 2018](#); [Di Pietro, 2012](#); [Horstschräer and Sprietsma, 2015](#); [Triventi et al., 2017](#)). Despite reshaping higher education for over 400 million Europeans, the reform's consequences for workers' earnings and careers have remained entirely unknown. Bologna created a natural experiment comparable in structure to what the US community college

literature exploits — expanding access to tertiary education by lowering its time cost (Mountjoy, 2022) — yet its labour market consequences represent a complete blind spot in the literature. Uniquely, however, Bologna also restructured the full degree ladder, making it possible to study how access expansion affects not just entry into higher education but continuation to advanced degrees. Second, I exploit data that enable a substantially richer analysis of earnings dynamics than has been possible in the access-expansion literature. Mountjoy (2022) and related work use administrative education records linked to unemployment insurance earnings registries, which track quarterly wages but contain no information on employers, firm quality, or job transitions. Quadros de Pessoa is a longitudinal employer-employee panel, which allows me to go well beyond average wage effects and document lifecycle earnings trajectories, wage decompositions separating human capital from firm sorting, and early-career progression within and across firms. Third, my findings on field reallocation contribute to the literature on heterogeneous returns to field of study (Bleemer and Mehta, 2022; Hastings et al., 2013; Kirkeboen et al., 2016). Whereas those papers identify marginal students through academic ability cutoffs, I identify students constrained by the time cost of completing a given field of study, and show that removing this constraint induces large reallocation away from healthcare and toward STEM and Economics.

The rest of the paper is structured as follows: section 2 describes the reform and institutional background, section 3 the data and section 4 the empirical strategy and identification, section 5 describes the main findings, section 6 provides suggestive evidence on plausibly low-SES individuals, and lastly section 7 concludes.

2 Reform background

The Bologna Process fundamentally represented a choice to converge European higher education toward the Anglo-Saxon model rather than maintaining the diverse Continental European traditions. Three key factors drove this decision. First, Anglo-Saxon countries demonstrated substantially higher tertiary education participation rates—by 2004, 39 percent of Americans aged 25-64 held tertiary degrees compared to just 23 percent of Europeans, suggesting the shorter degree structure successfully expanded access (Aghion, 2006). Second, the Continental European landscape had become increasingly fragmented, with degree lengths varying from four to six years across countries and fields, creating significant barriers to student mobility

and degree recognition. In contrast, the Anglo-Saxon three-year bachelor’s followed by an optional two-year master’s provided a transparent, standardized framework that facilitated cross-border movement of students and workers. Finally, the compressed timeline to a first degree reduced both the financial burden and opportunity cost of higher education, allowing students to enter the labour market sooner or pivot more easily if their initial field choice proved unsuitable—features particularly valuable for students from disadvantaged backgrounds who faced greater constraints in committing to lengthy programs (Jacobs and Van Der Ploeg, 2006).

2.1 Institutional Background in Portugal

Portugal’s implementation of this Anglo-Saxon convergence provides an exceptional natural experiment for studying the effects of compressing higher education degrees. Portugal maintained a binary system comprising universities, which focused on general academic education and research, and polytechnic schools, which were traditionally more vocationally oriented. Both sectors offered the licenciatura as the first degree, followed by optional postgraduate programs including the mestrado (master’s degree). The Bologna framework, initiated by the 1999 Bologna Declaration and reinforced through subsequent ministerial conferences in Prague (2001), Berlin (2003), and Bergen (2005), aimed to create a European Higher Education Area with comparable degree structures across countries. The core requirement was adopting a three-cycle system: a bachelor’s degree (typically three years), master’s (normally two years), and doctorate.

Field-specific restructuring. The reform’s impact varied substantially across fields of study, generating differential changes in the cost-benefit calculus of different degree choices that are central to interpreting the labour market effects documented below. Fields with direct access to regulated professions — including engineering, architecture, and medicine — were restructured as integrated masters of five years preserving the total time required for professional qualification. An integrated master is a single uninterrupted programme that combines the bachelor’s and master’s cycles into one continuous qualification, meaning students enrol directly into a five-year degree without first completing a standalone three-year bachelor’s and then separately applying for a master’s — and crucially, tuition fees are capped at the undergraduate rate for the entire duration, making it substantially cheaper than completing a three-year bachelor’s followed by a separately

funded two-year master's. However, these programs introduced a crucial new feature: students could exit after three years with a recognised bachelor's degree, even if this credential did not grant access to the regulated profession itself. This option value was absent under the previous system, where abandoning a five-year *licenciatura* mid-way yielded no formal qualification. The reform therefore reduced the downside risk of enrolling in STEM and professional fields substantially: the worst-case outcome for a student who enrolled but ultimately did not complete the full integrated master shifted from receiving no credential to receiving a three-year bachelor's degree with recognised labour market value. Fields whose graduates primarily entered non-regulated occupations — including economics, business, social sciences, and humanities — were restructured as three-year bachelor's programmes, with an optional two-year master's available for those seeking further specialisation or academic careers. For these fields, the reform represented a straightforward time compression: the pre-existing four-year *licenciatura* was shortened by one year while preserving access to the same labour market destinations. Given that economists and business graduates had documented positive wage premia in the Portuguese private sector, this reduction in time-to-degree made these fields unambiguously more attractive on a cost-benefit basis, particularly at the margin for students who were previously deterred by the four-year commitment. Nursing programs were not similarly compressed and retained a four-year structure in the post-reform system. This asymmetry arose because nursing qualifications are governed by separate European professional directives that mandated minimum hours of clinical training incompatible with a three-year structure, leaving Portuguese institutions with limited flexibility to shorten the program. The consequence was a deterioration in nursing's relative attractiveness: prior to the reform, nursing and most other subjects required four years, but after the reform the other subjects could be completed in three while nursing remained at four.

Implementation timeline. The Portuguese Ministry of Science, Technology and Higher Education (MCTES) implemented Bologna as part of a broader reorganisation of the higher education system, occurring against a backdrop of significant market pressures. The sector faced growing imbalances between supply and demand: while enrollment in public universities had increased by 62% during the 1990s, demographic changes subsequently reduced the candidate pool, admission standards tightened with the reintroduction of national exams and minimum grades, and capacity expanded through public sector investments. This

created an increasingly competitive environment where institutions needed to differentiate themselves to attract students. The reform’s implementation schedule offered institutions strategic flexibility that proved revealing. Universities could begin implementing Bologna-compliant curricula in the 2006/07 academic year, with mandatory completion by 2008/09, yet virtually all programs had converted by the 2007/08 academic year. As documented by [Cardoso et al. \(2008\)](#) in the reform’s first year, approximately 43% of study programs adapted to Bologna principles by 2006/07 and only one institution delayed the implementation of curricula changes to the 2008/09 academic year. For my empirical analysis, this implementation structure offers several advantages. First, the compressed timeline — with virtually all programs transitioning by 2007/08 — creates a sharp discontinuity rather than the gradual phase-in that complicates identification in other countries. Second, Portugal’s standard university entry age of 18 aligns perfectly with the reform timing: cohorts turning 18 in 2007 or later faced the new system when making enrollment decisions, while older cohorts had already entered or foregone university under the previous system. This age-based discontinuity, combined with Portugal’s comprehensive administrative data on education and employment, enables clean identification of the reform’s causal effects on both educational attainment and labour market outcomes. Importantly, the field-specific restructuring described above means that the reform did not simply shift all students toward shorter degrees: it changed the relative cost and option structure of different fields in ways that generated predictable patterns of field reallocation, which the empirical results in Section 5 confirm and which inform the interpretation of the wage and labour market effects.

3 Data

3.1 Quadros de Pessoal (QP)

Quadros de Pessoal (QP) is a rich and comprehensive linked employer-employee administrative data collected annually by Portugal’s Ministry of Employment. It is a panel dataset that encompasses all private sector establishments with at least one employee, with wage data recorded each October. While it excludes civil servants, the self-employed, and domestic workers, it captures the complete universe of workers and firms in manufacturing and private services.

The following variables are reported on each worker: gender, birth date, educational attainment, occupation, firm entry date, monthly compensation, working hours, and hierarchical position (such as manager or highly qualified employee). Educational variables capture both the highest degree completed and the field of study. Employer characteristics include firm size and geographic location. To construct hourly wages, I divide total monthly compensation—comprising base salary, seniority premiums, and other regular payments—by standard monthly hours worked. All wage measures are deflated to 2010 euros using the consumer price index. I use information stretching from 2000 to 2023. However, no worker data are available for 2001. Table 2 in the Appendix presents the main descriptive statistics.

I construct two important measures related to earnings and wages. As the reform comes in place in 2007, post-reform individuals will be observed latest at age 34, whereas pre-reform individuals are observed up to age 41. Similarly, post-reform individuals will be observed, on average, at later calendar years compared to pre-reform cohorts (as shown in Table 2 in the Appendix). To control for any time-specific trends that might confound wages, I index wages using older cohorts observed in the same calendar year and of the same gender. A more detailed description of the construction of relative wages can be found in Section A.1 in the Appendix. Secondly, I construct cumulative annual earnings by a given age. For each individual observed in the dataset, I set annual earnings to zero if the person has no observed earnings at a specific age, and I compute cumulative earnings at specific age A by aggregating all the cash flows earned up to age A . A more detailed description of the construction of cumulative earnings can be found in Section A.2 in the Appendix.

3.2 Labor Force Surveys (LFS)

The Labor Force Survey (LFS) provides a nationally representative sample of Portugal’s entire population, collected quarterly by the National Statistics Institute. Unlike the QP data, the LFS is a repeated cross-section rather than a panel—I observe different individuals in each survey wave and cannot track the same person over time. However, this dataset offers crucial advantages for benchmarking: it covers all employment statuses including public sector workers, the unemployed, and those out of the labour force, providing an unselected view of the population’s educational and employment outcomes. The survey records detailed information on educational attainment, field of study, employment status, occupation, and wages, using

similar educational classifications as the QP data. This compatibility allows me to validate that my QP-based results on educational attainment are not driven by selection into private sector employment. I use LFS data spanning the same period from 2000 to 2023, enabling consistent comparisons across both datasets throughout the pre- and post-reform periods. While the repeated cross-sectional nature prevents me from analyzing individual career trajectories, the LFS serves as an essential robustness check to ensure my findings reflect population-wide effects rather than compositional changes in private sector employment.

4 Empirical Strategy

Using the data described in the previous section, I construct a running variable $Z_i = c_i - 2007$, where $c_i = \text{Year of birth} + 18$, which captures the distance, in years, between the year individual i was 18 and the year in which the reform was fully implemented, i.e. 2007. To identify treated individuals, i.e. those more exposed to the reform, I use the treatment dummy $D_i = \mathbf{1}[Z_i \geq 0]$.

To estimate the impact of the reform, I estimate the following Fuzzy Regression Discontinuity equation:

$$Y_i = \alpha_0 + \alpha_1 D_i + \alpha_2 Z_i + \alpha_3 Z_i \times D_i + \alpha_4 \mathbf{X}_i + \eta_i \quad (1)$$

where α_1 is the coefficient of interest, α_2 is the slope of the relationship between the running variable and the outcome before the reform, for $D_i = 0$, and α_3 is the slope after the reform. $\mathbf{X}_i$ is a vector of controls including age polynomials. Y_i captures education and labour market outcomes. Education outcomes are a series of dummies taking value one if an individual's highest education is: (i) a university degree; (ii) a first-cycle university degree; (iii) a university degree in a given field. To proxy years of schooling, I use the age at which an individual reports to have started to work in the QP dataset while I use the age at which an individual reports leaving their highest education from the LFS. Labour market outcomes are: (i) cumulative annual earnings and relative hourly wages (raw and logs); (ii) worker and firm fixed effects as explained in the AKM section 4.1; (iii) dummies for whether the occupation matches the field of study; (iv) a dummy for whether the individual has a highly qualified position (manager, quadro, or highly qualified employee); (v) wage growth in first 3 years; (vi) firm premium growth in first 3 years.

Identification assumptions The main identification assumption on which my empirical strategy is based is that individuals cannot manipulate the running variable in order to benefit from the reform. As individuals do not have the power to change the year in which they were born, it is reasonable to assume that the density is smooth around the cutoff. To formally test this, I estimate separate regressions of the log density on each side of the cutoff and test for a jump at the cutoff, $Z_i = 0$. Specifically, I compute the density in each cohort bin as $\hat{f}(z_j) = N_j / (N \cdot h)$ where $h = 1$ cohort year is the bin width, and run

$$\log \hat{f}(z_j) = \alpha_L + \beta_L z_j + u_j \quad (z_j < 0), \quad \log \hat{f}(z_j) = \alpha_R + \beta_R z_j + v_j \quad (z_j \geq 0), \quad (2)$$

using weights N_j (bin counts) and a triangular kernel within a symmetric window around the cutoff. The parameter of interest is the discontinuity in log density, $\Delta = \alpha_R - \alpha_L$, with null $H_0 : \Delta = 0$. I report the point estimate $\hat{\Delta}$, its standard error, and p-value. Results for this exercise are reported in Figure 1, where there is no visual break in the density at the cutoff. A formal test finds that the change in the log density around the cutoff is not statistically significant.

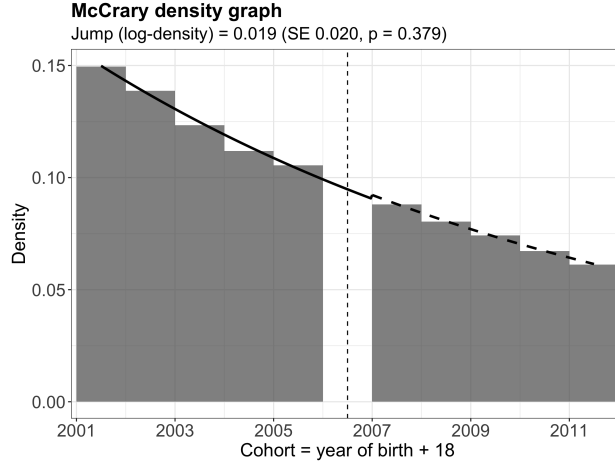


Figure 1: McCrary density test for manipulation of the running variable

Importantly, however, individuals have the power to decide at what age to enroll into university, and cohorts who were 18 years old before the reform was implemented could actively choose to delay their university enrollment in order to benefit from shorter degrees. This would be particularly relevant if individuals could anticipate the reform implementation. I test this assumption by estimating equation (1) using as an outcome the share of individuals that are observed to be working between graduating from high-school and

graduating from university. Intuitively, if we were to see an increase in the share of people who take up a job before going to university, this would suggest that the reform likely delayed the timing of university enrollment. Results for this test are reported in Table 3 in the Appendix, where we see that the reform has no effect on the share of people who work between graduating high-school and graduating university, and that this is true for both the female and male sample.

4.1 AKM model for firm sorting analysis

To examine whether the reform changes worker quality and worker sorting across firms, I implement the Abowd, Kramarz, and Margolis (AKM) model (Abowd et al., 1999). This approach decomposes log wages into worker characteristics, firm-specific pay premiums, and residual variation. The AKM methodology requires restricting the analysis to the largest connected set of firms—that is, firms that are linked through worker mobility. Two firms are connected if at least one worker moves between them during the sample period, and by extension, all firms form a connected set if they can be linked through chains of such moves. This restriction ensures that worker and firm effects are separately identified through the mobility patterns in the data. In Portugal’s relatively integrated labour market, the largest connected set encompasses the vast majority of firms and workers.

By including firm and worker fixed effects I am able to control for unobservables at the firm and worker level that capture a substantial amount of wage variation, while mitigating potential endogeneity problems. More specifically, I estimate the following equation:

$$y_{it} = X_{it}\beta + \alpha_i + \theta_j + \mu_t + \epsilon_{it} \quad (3)$$

where y_{it} is the log hourly wage for worker i in year t , X_{it} includes a quadratic on the experience and tenure of the worker, α_i are worker fixed effects, θ_j represents a fixed effect for the firm j , and μ_t are year fixed effects. By tracking changes in the average firm and worker fixed effect for workers before and after the reform, I can determine whether treated cohorts earn more because they sort into higher-paying firms or because they are, on average, better workers.

4.1.1 Decomposing wage effects

To interpret the reform’s impact on wages through the lenses of the AKM model, I decompose the discontinuity in log wages at the cutoff into the contributions of worker composition, firm sorting, and a remainder term. After estimating equation (3) once on the whole sample to obtain a single, common set of worker effects $\hat{\alpha}_i$ and firm effects $\hat{\theta}_{f(i,t)}$ under a connected set, I construct, for each observation, the identity $y_{it} = \hat{\alpha}_i + \hat{\theta}_{f(i,t)} + (X_{it}\hat{\beta} + \hat{\varepsilon}_{it})$. I then run the same RD regression as specified in equation (1) separately on y_{it} , on $\hat{\alpha}_i$, on $\hat{\theta}_{f(i,t)}$, and on the remainder $Other := y_{it} - \hat{\alpha}_i - \hat{\theta}_{f(i,t)}$ (which aggregates $X_{it}\hat{\beta}$ and the residual). By linearity of OLS and because all components are analyzed on the same sample with the same right-hand side, the estimated jump in wages satisfies

$$\Delta y = \Delta \bar{\alpha} + \Delta \bar{\theta} + \Delta(\overline{X\hat{\beta}}) + \Delta \bar{\varepsilon} \quad (4)$$

up to rounding. In this decomposition, $\Delta \bar{\alpha}$ captures changes in the average worker effect, $\Delta \bar{\theta}$ captures changes in the average firm pay premium (sorting to higher- or lower-paying firms), and the “Other” term captures changes in observables and unexplained residual variation. I report results for the whole sample and by gender (using the same $\hat{\alpha}_i$ and $\hat{\theta}_{f(i,t)}$ to preserve additivity) and present each component’s share of the total wage jump ($\Delta \text{component} / \Delta y$).

5 Results

5.1 Education

Figure 2 shows how the reform affected the probability of graduating university and the average time spent in university - by which someone who did not go to university spends 0 years in university. First, we observe that females have higher levels of education than males at baseline. Secondly, we observe that the reform substantially changes educational outcomes. For both females and males, the probability of graduating from university increases by roughly 8-9%. Overall, there is a positive effect on schooling, on net: average years spent in university increase. However, this is stronger for the female sample. Table 1 presents the main results on educational attainment using LFS and QP data. The Bologna reform increased overall university graduation rates by 1.8 percentage points in the LFS sample and 2.1 percentage points in

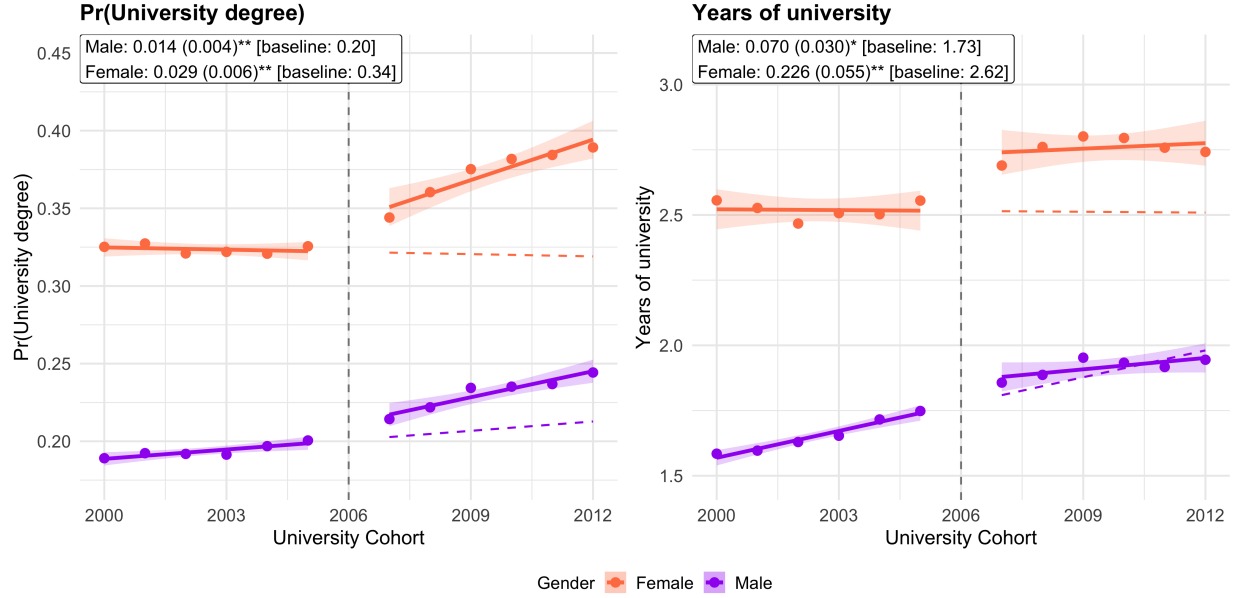


Figure 2: RD graphs on highest education and years of university

the QP sample. As expected, the baseline share of people completing a university degree is lower in the LFS sample (18% vs 24%). In both datasets, the percentage point increase is stronger for the female sample. The second outcome shows the effects of the reform on the probability of having completed a first-cycle university degree (bachelor's) as highest education. In the LFS sample, this increase outweighs the overall increase in university attainment (2.3pp vs 1.8pp), indicating that not only the reform operated almost entirely through the extensive margin, but also that the share of people completing a master's degree declined and thus that the reform also operated on the intensive margin. In the QP sample, this pattern can be observed for the male sample only - a 2pp increase in bachelor's attainment compared to a 1.5pp increase in overall university attainment. Females in QP data instead show a 2.2pp increase in bachelor's attainment compared to a 2.8pp increase in overall university attainment. The last outcome of Table 1 shows the effects on the age in which individuals report finishing their highest education (LFS data) or the age in which individuals report starting to work (QP) as a proxy for years of schooling. In both cases, we see a positive net effect on this years of schooling proxy, indicating that the reform likely increased overall educational attainment in the population. Overall, both datasets highlight a strong extensive margin effect of the reform, as well as heterogeneity by gender, suggesting that overall schooling in the population increased with the reform. The

Table 1: Education outcomes: LFS vs QP

	Probability that highest education is:				Age when:	
	Any university		First-cycle uni		End Educ	Start Work
	LFS	QP	LFS	QP	LFS	QP
All: $N_{QP} = 12,648,491$; $N_{LFS} = 310,355$						
RD Jump	0.018*** (0.003)	0.021*** (0.001)	0.023*** (0.003)	0.021*** (0.001)	0.254*** (0.032)	0.142*** (0.005)
Baseline	0.180	0.244	0.123	0.204	18.066	21.997
Female: $N_{QP} = 6,025,535$; $N_{LFS}=153,626$						
RD Jump	0.027*** (0.005)	0.028*** (0.001)	0.032*** (0.004)	0.022*** (0.001)	0.244*** (0.044)	0.216*** (0.007)
Baseline	0.226	0.311	0.157	0.265	18.395	22.067
Male: $N_{QP} = 6,622,956$; $N_{LFS}=156,729$						
RD Jump	0.008** (0.004)	0.015*** (0.001)	0.015*** (0.003)	0.020*** (0.001)	0.264*** (0.045)	0.073*** (0.008)
Baseline	0.135	0.183	0.089	0.148	17.741	1.549
p-val (F=M)	0.002	0.000	0.003	0.057	0.747	0.000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Notes: Entries show donut RD jumps at the 2007 cohort (excluding 2006), with linear trends in the running variable fitted separately pre/post. Controls include age polynomials. “Baseline” is the mean in the cohort immediately before the reform. LFS = Labour Force Survey; QP = Quadros de Pessoal. The first dependent variable is the probability of an individual’s highest education being a university degree. The second one is the probability of an individual’s highest education being a first-cycle degree. The third outcome is **age at completion of highest education** for LFS and **age started working** in the private sector for QP.

QP data, however, seems to show stronger effects on master’s completion for females, highlighting a possible over-representation of female master graduates compared to the whole population.

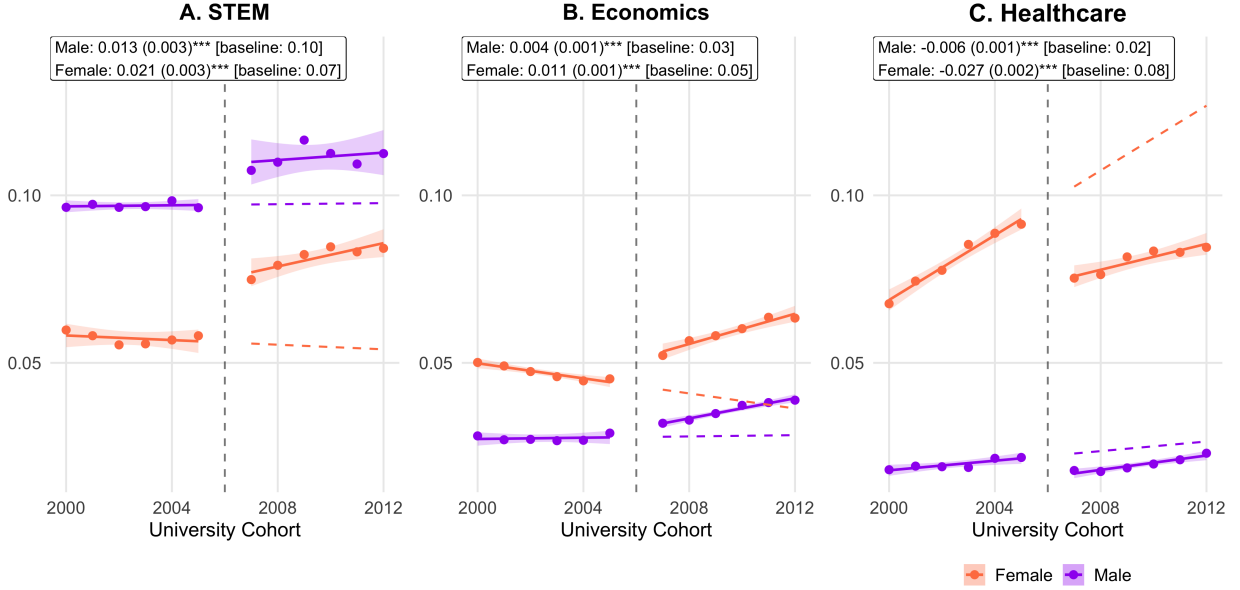


Figure 3: RD graphs on field of study

Field of study reallocation Figures 3 and 4 document substantial reallocation across fields of study induced by the reform. The outcomes plotted are unconditional attainment rates — the share of workers in a given birth cohort who hold a degree in each field — rather than field choice conditional on university enrollment. This distinction matters for interpretation: a positive RD estimate reflects both the extensive margin (more people obtaining a degree at all) and the intensive margin (field switching among those who would have enrolled regardless), and cannot be attributed solely to changes in the composition of who chooses which subject. The most striking pattern is a shift away from healthcare/nursing programs and towards STEM and economics. Healthcare/nursing programs experienced a 2.7pp decline (33% relative to baseline) for females and a 0.6pp decline for males (30% relative to baseline). This contraction is naturally explained by a deterioration in the relative attractiveness of nursing under the new system: while most fields were shortened to three years, nursing retained a four-year structure post-reform. Students comparing programmes therefore faced a world in which the opportunity cost of nursing — in terms of years foregone relative to alternatives — increased substantially, reducing its appeal at the margin.

The expansion into STEM, where females are 2.1pp more likely to obtain a degree (30% relative to baseline) and males 1.3pp more likely, operates through a different channel: option value. Under the Bologna

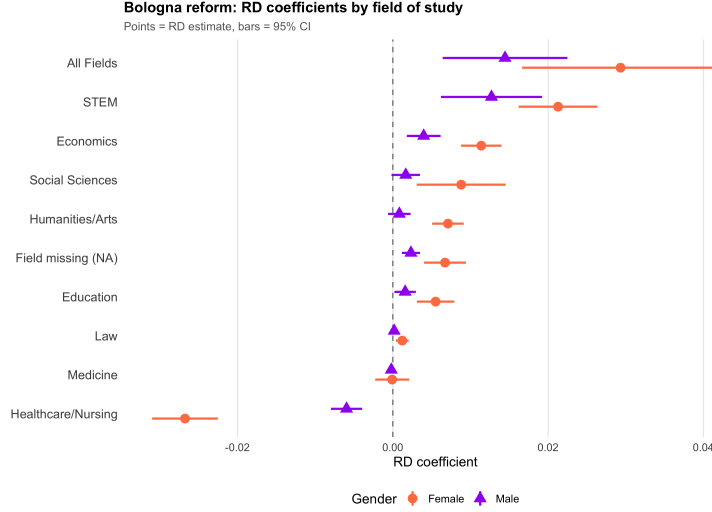


Figure 4: RD estimates on field of study

structure, STEM programmes follow a 3+2 format in which entry to the regulated profession still requires the full five years as before, but students now have the option to exit after three years with a recognised bachelor's degree. This reduction in downside risk — the worst-case outcome is a three-year degree rather than no credential — makes STEM more attractive to students who were previously deterred by the length and uncertainty of the commitment. Economics, meanwhile, benefits from a straightforward time compression effect: the degree was shortened from four to three years while maintaining access to a labour market with high and well-documented returns, making it unambiguously more attractive on a cost-benefit basis. Table 4 in the Appendix shows the comparison between LFS and QP samples. To maintain comparability across the samples due to different classifications of fields of study, I aggregated in the QP data: (i) economics with social sciences and law, and (ii) healthcare with medicine. We see very similar patterns in the composition of the fields of study using LFS and QP data, i.e. an increase in STEM and economics and social sciences as well as a decline in healthcare; in both datasets, the effects are stronger for the female sample.

5.2 Cumulative Earnings

The challenge of capturing full welfare effects is that the reform operates on both the extensive and the intensive margin of university attendance. On the extensive margin, more students pursuing university delays labour-market entry (reducing early cumulative earnings through forgone wages) but potentially increases

later earnings through higher human capital. On the intensive margin, students exiting after shorter degrees gain additional years of work experience by any given age (boosting cumulative earnings through more years in the labour force) but may sacrifice human capital from foregone education (potentially reducing their wage levels). Cumulative earnings by a given age captures both these timing effects (when people enter the labour market) and human capital effects (what they earn once there), providing a net welfare measure that integrates all mechanisms through which the reform operates. Rather than attempting to decompose and separately identify each channel through which the reform operates, cumulative earnings measures the net effect: the total income generated by treated cohorts from labour market entry through a given age, compared to control cohorts over the same lifecycle period. This approach naturally incorporates both the costs and benefits of the reform. Years spent in additional education appear as foregone earnings (zeros or low values during schooling), while subsequent returns to human capital are captured through the full earnings trajectory post-graduation.

Figure 5 presents the RD coefficients for cumulative earnings across ages. The results reveal a striking

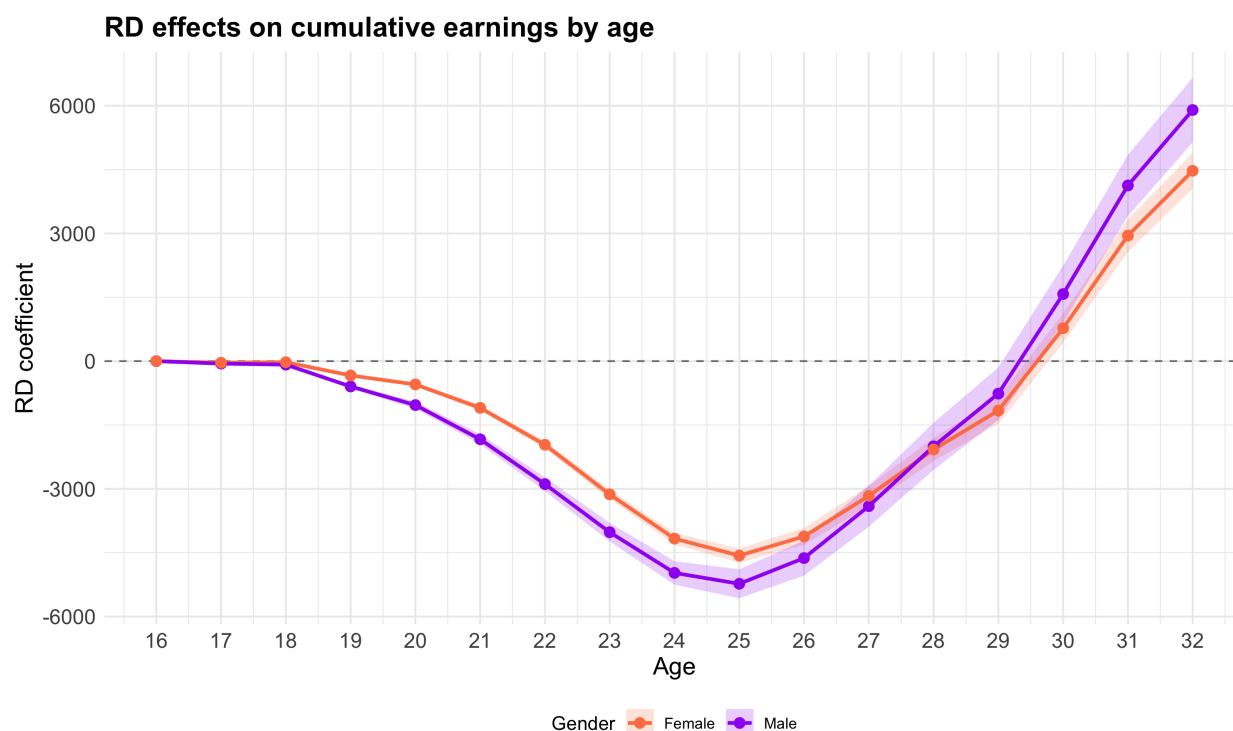


Figure 5: RD coefficients for the effect of the reform on cumulative earnings across ages, constructed as outlined in Section A.2.

intertemporal pattern consistent with human capital investment dynamics. In the early 20s the effect of the reform is negative: more people go to university and hence they forego earnings. In the mid 20s individuals start to finish their studies and enter the labour market, with a higher education level as their pre-reform counterparts, and hence start catching up on the foregone earnings. By the early 30s, the effect on cumulative earnings turns positive, i.e. the initial losses from foregone earnings while studying are completely offset by higher wages once individuals start working. The next section will focus on hourly wages specifically.

5.3 Earnings, firm and worker quality

Returns to higher education The first column of Table 5 shows that hourly wages increase by 1.6% for the whole sample, with a larger effect for females (2.1%) than males (1.3%), a difference that is statistically significant. A simple back-of-the-envelope calculation merging education and wage results provides useful context: dividing the 1.6% wage increase by the 0.14-year increase in schooling (proxied by delayed labour market entry) in Table 1 yields an implied return of 11.4% per year of education, broadly consistent with existing estimates for Portugal. For instance, [Portugal et al. \(2024\)](#) find a 9.8% return per year of education using changes in compulsory schooling.

Decomposing wage effects into worker and firm channels A natural first question is whether these wage gains reflect improvements in workers’ own productivity and human capital, or whether they are driven by compliers sorting into better-paying employers. Table 5 addresses this by decomposing the overall wage RD estimate into the contribution of firm fixed effects — capturing the component of wages attributable to the employer rather than the worker — and the residual non-firm component, following the AKM decomposition in equation (4). The results reveal that approximately 30% of the total wage effect operates through the workplace channel: compliers sort into firms that pay systematically higher wages to all their employees, above and beyond what individual productivity would predict. The remaining 70% reflects improvements in worker-level outcomes — human capital accumulation, credentialling, and field reallocation toward higher-return occupations. This decomposition is stable across genders, with the firm channel accounting for 31.6% of the female wage gain and 26.4% of the male gain, despite the overall wage effects differing significantly in magnitude. The similarity in the firm channel share across genders — despite the absolute size of the

firm FE effect being larger for women — implies that the mechanisms generating firm sorting are broadly common to both groups, even if the magnitude differs.

Is the firm sorting advantage sustained or a one-time match? The level decomposition in Table 5 establishes that compliers work at better firms, but it does not reveal whether this advantage arises at labour market entry and is subsequently maintained, or whether it accumulates over time through active job-to-job mobility toward higher-paying employers. These two mechanisms have very different implications: the first points to a signalling or credentialling story in which the degree improves the initial match, while the second points to a search story in which graduates leverage their qualification to move up the job ladder over time. Table 7 helps distinguish between these channels by examining both job mobility and wage growth over the first three years of the career. The first column shows that, pooling both genders, treated cohorts are 1 percentage point *less* likely to change firms in the first three years of employment — a finding that directly contradicts the search channel hypothesis. If the firm sorting advantage were driven by graduates using their credential to search more effectively for better employers, one would expect higher, not lower, mobility. The reduction in mobility is concentrated among men, who are 1.5 percentage points less likely to switch firms, while the female effect is smaller and statistically insignificant. This suggests that male compliers in particular enter firms that are good enough matches that they have less reason to search, while female compliers show more mixed mobility patterns. Despite switching firms less frequently, treated cohorts experience higher wage growth regardless of whether they stay or move: the second and third columns of Table 7 show wage growth increases of 2.1 percentage points for both stayers and movers in the whole sample. For women, stayers experience slightly larger wage growth (2.8pp) than movers (2.5pp), while for men both groups show more modest gains (1.5pp and 1.8pp respectively), with no statistically significant difference between them. The near-identical wage growth for stayers and movers establishes that the Bologna wage advantage does not depend on job switching — it materialises within the entry firm as well as across firms, and is therefore not a selection story in which only those who successfully move to better employers benefit.

Dynamic decomposition: how the firm channel evolves Table 6 extends the level decomposition to wage *growth* over the first three years, asking what share of the dynamic wage advantage flows through the

firm channel. The results reveal a striking gender asymmetry that was invisible in the level decomposition. For the whole sample, the firm channel accounts for only 5.8% of wage growth — compared to 30% of the wage level effect — indicating that the firm sorting advantage is primarily established at entry rather than accumulating through subsequent mobility toward better employers. However, this aggregate masks sharply divergent patterns by gender. For women, 14.6% of wage growth operates through the firm channel, consistent with female treated cohorts actively sorting into higher-paying employers over the early career even conditional on their entry firm. For men, the firm channel contribution to wage growth is negative, meaning that male treated cohorts' employers actually decline in relative quality over the first three years, even as their overall wages grow. The male wage growth advantage is therefore entirely driven by non-firm factors — within-firm wage progression, job title changes, and other components of the individual wage — rather than by improvements in employer quality. This decomposition, combined with the mobility evidence, paints a

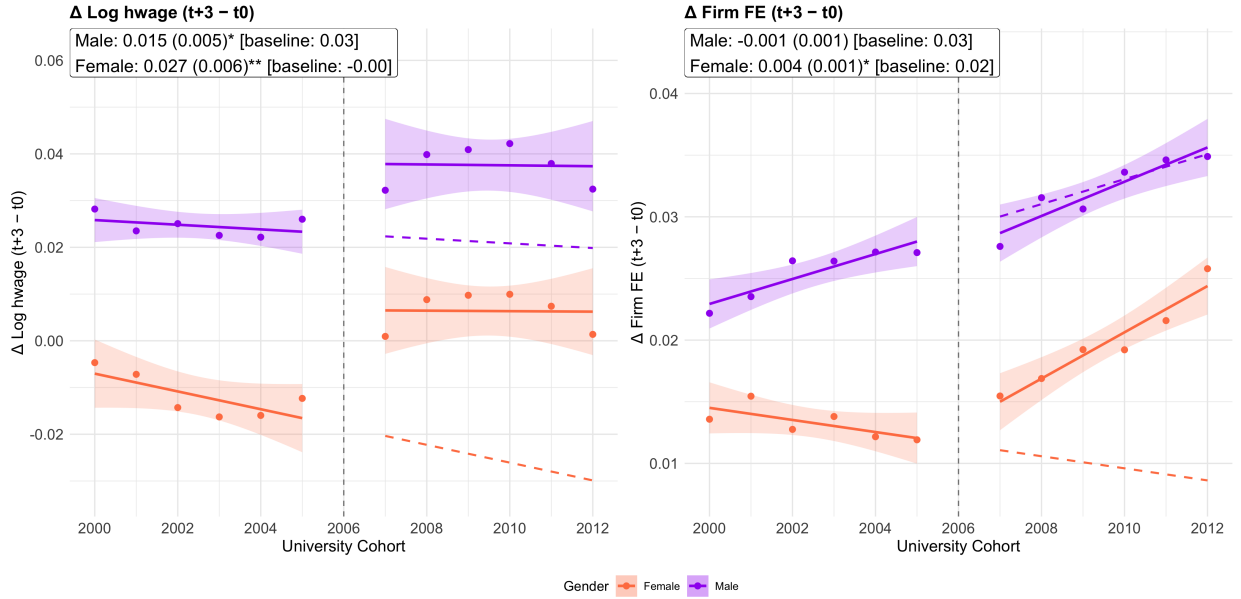


Figure 6: Early career changes in wages and firm premium. Firm FE as estimates as outlined in equation (3).

coherent picture of how the Bologna reform generated its wage effects. Both male and female treated cohorts sort into better-paying firms at labour market entry — the 30% firm channel in the level decomposition — and this initial advantage is sustained by lower job mobility, as workers who entered good matches have less incentive to search. However, the subsequent wage trajectories diverge: female treated cohorts continue to

improve their employer quality over time and accumulate wage growth both within and across firms, while male treated cohorts' wage growth comes entirely from within their entry firm, with no improvement — and indeed a slight deterioration — in the firm quality component. The degree therefore functions primarily as a signal that improves the initial job match for both genders, but additionally enables ongoing employer upgrading for women in ways that do not materialise for men.

6 Suggestive evidence on student types

In this section, I explore two approaches aimed at disentangling the impact of the Bologna reform on two distinct groups: (i) individuals who would have enrolled in university regardless of the reform, and (ii) individuals whose university enrollment is induced by the reform. The main difficulty is that the reform changes both who goes to university and the outcomes they experience once they get a degree. As a result, it is hard to cleanly separate effects. The goal of this section is therefore to provide suggestive evidence on how the reform's impacts differ across these margins.

Approach 1: SES Proxies The first approach uses observable characteristics to identify subsamples that are likely coming from low-SES families. I focus on individuals who are observed entering the labor market before age 18. Entering work before age 18 implies that these individuals started working before completing upper-secondary education, and therefore are likely to come from lower socio-economic backgrounds and to be less academically oriented. For this group, the Bologna reform—by shortening time to degree and reducing the overall cost of university—is plausibly more salient, making enrollment more responsive to the reform. As a result, estimated effects within this subsample are likely to be driven primarily by compliers. While this approach does not perfectly isolate compliers, it provides a simple and intuitive way to study reform effects among individuals at the margin of university participation.

To validate this proxy, I verify that the share of individuals classified as low-SES does not change discontinuously at the reform threshold. A discrete jump would suggest that the proxy is itself affected by the reform—for instance, which would invalidate its use as a pre-determined characteristic. Reassuringly, I find no significant discontinuity at the reform cutoff, suggesting that the proxy captures a stable underlying

”type” rather than an outcome of treatment.

Approach 2: Lee Bounds The second approach uses Lee bounds to provide a range of plausible effects for always-takers without requiring proxy variables. The key idea is that the Bologna reform increased university enrollment, so some of the post-reform graduates are compliers who would not have enrolled otherwise. Under the assumption that the reform weakly increases enrollment, these compliers must come from either the top or the bottom of the wage distribution among graduates. I implement this approach by trimming the post-reform wage distribution by an amount equal to the estimated increase in university enrollment caused by the reform. Trimming from the bottom gives an upper bound on the returns for always-takers, while trimming from the top gives a lower bound. Together, these bounds provide a range of plausible effects for individuals who would have gone to university regardless of the reform.

6.1 Lower-SES subsample

The results in this subsection focus on a group of individuals that serves as a proxy for low socio-economic status. Specifically, this subsample consists of individuals who are observed entering the labor market before age 18, which suggests that they started working before completing upper-secondary education and are therefore likely to come from disadvantaged backgrounds. This group includes roughly 70,000 individuals, compared to about 1.8 million individuals in the full sample. Despite its smaller size, the results for this low-SES proxy group reveal several striking differences relative to the full population, and are summarized. First, Panel (a) of figure 9 shows that the probability of obtaining a university degree increases by around 62% for women, rising from 4% before the reform to 4.25% after. Individuals in this group are much less likely to attend university to begin with—pre-reform university attainment is 4% for females in this sample, compared to about 30% for females in the full sample—and therefore have more scope to respond at the extensive margin. At the same time, the magnitude of the effect suggests that the Bologna reform had a particularly strong impact on individuals at the margin of university participation. Second, a large share of the increase in university attainment in this group is driven by economics degrees, as shown in panel (d). For females, the probability of studying economics rises from 1% to 1.8%, accounting for roughly a third of the total increase in university enrollment in this subsample. This indicates that economics plays a central role

in explaining how marginal students respond to the reform. Meanwhile, the effects on other fields of study are completely absent (panels b and c). Finally, labor market outcomes improve substantially, as shown in panels (f) and (e). Hourly wages increase by roughly 3.6% for females, with 38% of this effect driven by sorting into higher-paying firms.

6.2 Bounding Effects for Always takers

The population-level wage gains documented above cannot directly identify how the reform affected returns to university education. The 6% increase in graduation rates means that post-reform university graduates are a different, larger group than pre-reform graduates, creating a fundamental selection problem. The marginal students—those induced to attend university only because of the shorter, more accessible degrees—may differ systematically from always-takers in both observable and unobservable characteristics that affect wages. To bound the effect on returns to education, I employ the trimming procedure developed by Lee (2009), which provides sharp bounds on treatment effects in the presence of sample selection. The key insight is that while we cannot identify which specific individuals are marginal students, we know their exact proportion: the reform increased graduation rates by approximately 6% of post-reform graduates (1.5/25.4) would not have attended university absent the reform. The Lee bounds approach makes no assumptions about the selection process beyond monotonicity—that the reform only induced students into university, never out of it. Instead, it considers two extreme scenarios that bound the true effect:

- **Lower Bound:** Assumes marginal students are positively selected—they would have earned the highest wages among university graduates had they attended pre-reform. To construct this bound, I trim the top 6% of the post-reform graduate wage distribution and compare the remaining distribution to pre-reform graduates. This provides the most conservative estimate, as it assumes the reform attracted exceptionally high-ability students who happened to be constrained by the longer degrees.
- **Upper Bound:** Assumes marginal students are negatively selected—they would have earned the lowest wages among graduates. I trim the bottom 6% of the post-reform distribution, providing the most optimistic estimate under the assumption that the reform primarily attracted lower-ability students previously excluded by the time and financial costs.

The true effect on incumbent students—those who would have attended university regardless—must lie within this range. If even the lower bound remains close to zero or positive, it would indicate that returns to university education were maintained despite the expansion in supply and compression in study time.

Figure 7 presents the Lee bounds on returns to university education across the wage distribution, tracking graduates from age 24 through 30. The results reveal a dynamic pattern that varies across the distribution. Across all quartiles, the actual wage effect for university graduates is substantially negative in the immediate post-graduation period, ranging from -8% at the lower quartile to -15% at the upper quartile at age 24. Even the upper bounds remain negative or near zero at these early ages, indicating that this initial wage penalty is not driven by selection. This suggests that employers were initially skeptical of the new three-year degrees, possibly viewing them as providing insufficient training compared to the traditional longer programs. The wage trajectories show marked improvement with labour market experience. By age 30, the

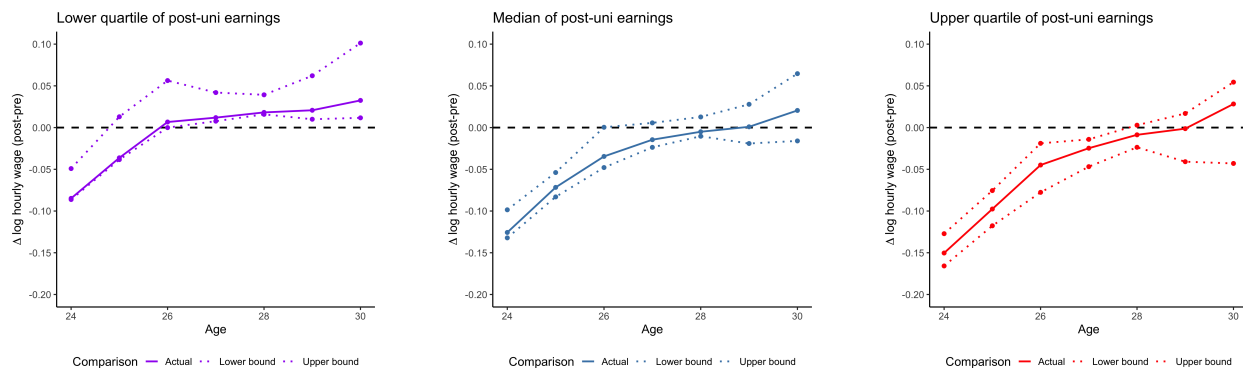


Figure 7: Lee Bounds on returns to university

actual effects converge to approximately 2-3% across all quartiles of the distribution. For graduates in the lower quartile, returns become unambiguously positive by age 28-29, with even the lower bound exceeding zero. By age 30, these workers experience 2-3% higher wages, with tight bounds suggesting minimal selection concerns. Notably, the lower bounds at age 30 remain close to zero or slightly positive across the distribution, while upper bounds reach 5-10%. This indicates that even under the most conservative assumptions about selection, the reform did not permanently reduce returns to university education.

The recovery is remarkably uniform across the wage distribution. While higher earners (upper quartile) experience the deepest initial penalty, they also show the steepest recovery trajectory. By age 30, the bounds

are relatively tight across all quartiles, suggesting that the true effect is close to the actual estimate of a small positive return. These patterns are consistent with a market learning story: employers initially discounted the new credentials but revised their beliefs upward as Bologna graduates demonstrated their productivity through actual work performance. The fact that returns fully recover and even turn positive by age 30 suggests that the shorter degrees ultimately maintained their value despite the compression in study time and the expansion in graduate supply. This finding is in line with previous results showing that the ratio of wages to years of schooling remains unchanged with the reform.

7 Conclusion

This paper studies how students respond when university degrees become shorter, using Portugal’s implementation of the Bologna Process as a natural experiment. By sharply reducing the time required to obtain a university degree, the reform changed both the cost of higher education and who finds university worth pursuing. In theory, such a reform could go either way: shorter degrees may attract new students and raise overall attainment, but they might also reduce human capital accumulation for those who would have completed longer programs. The evidence in this paper suggests that, on balance, the “less is more” hypothesis holds.

On the education side, the reform increased university graduation rates and raised average years of schooling in the population, indicating a strong response on the extensive margin. Shorter degrees made university accessible to students who would otherwise not have enrolled. At the same time, the reform led to sizable reallocation across fields of study. Students shifted away from healthcare and toward STEM and economics, which became more attractive under the new three-year structure. These field shifts matter for later outcomes and are particularly pronounced among women.

Labor market outcomes initially worsen, as more individuals stay in education longer and delay labor market entry. However, these early losses are temporary. By the early thirties, treated cohorts have higher cumulative earnings than comparable pre-reform cohorts. Hourly wages increase by about 4% on average, and decomposing wage effects shows that most of this gain reflects improvements in worker quality, with a smaller but meaningful role for sorting into higher-paying firms. Women, in particular, experience faster

early-career wage growth and stronger improvements in firm sorting.

Using two complementary approaches, I provide suggestive evidence that the reform improved outcomes for individuals at the margin of going and not going to university, as well as for individuals who would have likely gone anyways. Among individuals from likely disadvantaged backgrounds—proxied by early labor market entry—the increase in university completion is much larger than in the full population and is driven disproportionately by economics degrees. For this group, the reform leads to sizable labor market gains, with wage increases largely reflecting improved worker quality from obtaining a degree. Lee bounds instead confirm that always-takers are not permanently harmed. By trimming the top and the bottom of the wage distribution by the amount of “excess” graduates post-reform, I show that although graduates experience an initial wage penalty early in their careers, these differences largely disappear by around age 30.

Overall, the results suggest that shortening university degrees can expand access and improve long-run outcomes without undermining the value of higher education. While the reform involved short-run adjustment costs, especially in the early labor market years, these costs are outweighed by longer-run gains, particularly for students at the margin of university participation. These findings are relevant for policymakers considering reforms aimed at reducing the time and cost of higher education, and highlight the importance of maintaining flexibility and labor market recognition when implementing shorter degree structures.

Appendix

A Constructing Relative Wages and Cumulative Earnings

A.1 Relative Wages

To address the concerns on age and calendar time changing across birth cohorts, I construct relative wages that net out year-specific effects. I use the 1987-1999 cohorts as a comparison group, as these individuals were already 26-38 years old when Bologna was implemented and thus unaffected by the reform. For each individual in my cohorts of interest (2000-2012), I compute their wage relative to the average wage of older individuals observed in the same calendar year and of the same gender, and hence subject to the same macro-level shocks. For example, consider measuring wages at age 30:

- An individual from the 2000 cohort reaches age 30 in 2012
- Individuals from the 1987 cohort are age 43 in that same year (2012)
- I express the 30-year-old's wage as a ratio to the average wage of 43-year-old workers observed in the same calendar year and of the same gender

This is graphically shown in Figure 8.

This approach effectively controls for any gender and year-specific shocks that affect all workers regardless of their Bologna exposure, including business cycle effects, general wage trends, or other contemporaneous policy changes. The identifying assumption is that year-specific shocks affect different age groups proportionally within the same calendar year and gender, a reasonable assumption given that I compare prime-age workers (30-year-olds with 43-year-olds in the example above).

A.2 Cumulative earnings

For each individual i and age a , let y_{ia} denote real annual labour earnings (deflated to a common base year); we set $y_{ia} = 0$ when the person has no observed earnings (e.g., studying). We first pad the panel to one row per (i, a) over a pre-specified age grid (e.g., from school-leaving age up to $A \in \{25, 27, 30\}$) so that years

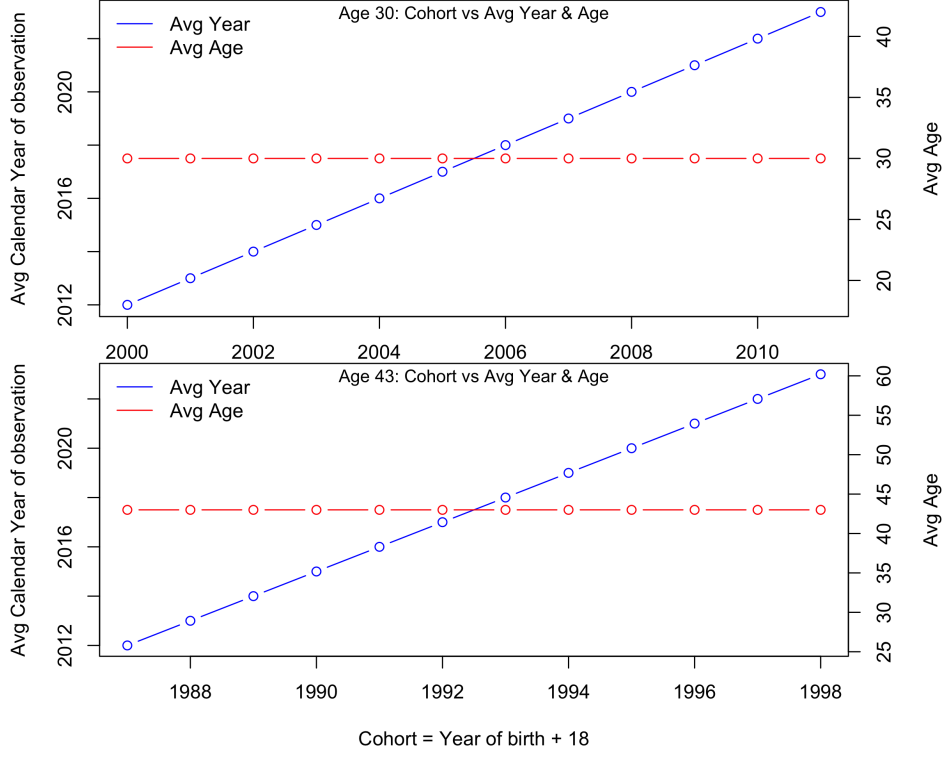


Figure 8: Explaining the construction of relative wages for individuals aged 30

outside employment contribute zeros rather than missing values. We then compute cumulative earnings in levels without discounting as

$$CE_i(A) = \sum_{a=a_0}^A y_{ia}$$

which aggregates all cash flows earned up to age A . To obtain a logarithmic outcome that is defined at zero and less sensitive to extreme values, we use the transformation

$$LCE_i(A) = \log(1 + CE_i(A)).$$

This construction treats forgone earnings while out of the labour market as true zeros (capturing delayed entry on the extensive margin) and additional earnings from earlier entry as positive contributions (capturing intensive-margin shifts), while avoiding post-treatment conditioning on employment or experience.

Table 2: Descriptive Statistics on QP data

VARIABLE	Control			Treatment		
	N = 8,074,453			N = 4,574,038		
	mean	min	max	mean	min	max
Age	30	17	41	27	17	34
Year	2014	2000	2023	2018	2006	2023
Female	0.48	0.00	1.00	0.47	0.00	1.00
University (any)	0.25	0.00	1.00	0.26	0.00	1.00
University (first-cycle)	0.20	0.00	1.00	0.18	0.00	1.00
Hourly Wage	5.13	0.07	1036	5.93	0.13	1790
Age at first Observation	22.66	17.00	41.00	22.19	17.00	41.00

Table 3: Testing the identification assumption on the reform affecting the decision of when to enroll in university

Pr(employed between graduating high-school and graduating university)	
	Whole sample
RD Jump	0.001 (0.001)
Baseline	0.105
	Female
RD Jump	0.002 (0.002)
Baseline	0.128
	Male
RD Jump	-0.000 (0.002)
Baseline	0.085
p-value Female = Male	0.283

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Notes: Entries show donut RD jumps at the 2007 cohort (excluding 2006), with linear trends in the running variable fitted separately pre/post. Controls include age polynomials. “Baseline” is the mean in the cohort immediately before the reform.

Table 4: Field of study results: LFS vs QP

	Probability that highest education is a uni degree in:							
	Health		STEM		LawSocSci		Other	
	LFS	QP	LFS	QP	LFS	QP	LFS	QP
Whole sample								
RD Jump	-0.008*** (0.001)	-0.015*** (0.000)	0.004** (0.002)	0.012*** (0.000)	0.011*** (0.002)	0.006*** (0.000)	0.011*** (0.002)	0.011*** (0.000)
Baseline	0.037	0.043	0.038	0.054	0.042	0.056	0.038	0.083
Female								
RD Jump	-0.014*** (0.003)	-0.025*** (0.001)	0.006*** (0.002)	0.014*** (0.000)	0.016*** (0.003)	0.012*** (0.001)	0.019*** (0.003)	0.019*** (0.001)
Baseline	0.062	0.073	0.027	0.035	0.060	0.079	0.049	0.109
Male								
RD Jump	-0.003** (0.001)	-0.005*** (0.000)	0.001 (0.002)	0.010*** (0.001)	0.007*** (0.002)	0.002*** (0.000)	0.005*** (0.002)	0.003*** (0.000)
Baseline	0.013	0.016	0.049	0.071	0.025	0.036	0.027	0.059
p-value Female = Male	0.002	0.000	0.077	0.000	0.013	0.000	0.000	0.000

* p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses.

Notes: Entries show donut RD jumps at the 2007 cohort (excluding 2006), with linear trends in the running variable fitted separately pre/post. Controls include age polynomials. “Baseline” is the mean in the cohort immediately before the reform. LFS = Labour Force Survey; QP = Quadros de Pessoal. The first dependent variable is the probability of an individual’s highest education being a university degree in Healthcare. The second one is the probability of an individual’s highest education being a university degree in STEM. The third one is the probability of an individual’s highest education being a university degree in law and social sciences. The fourth one is the probability of an individual’s highest education being a university degree in other fields.

Table 5: Decomposing wage effects into firm and non-firm AKM
fixed effects

	RD Jump		Share of wage effect (%)	
	Log(hwage)	Firm FE	Firm FE	Non-Firm FE
Whole sample				
RD Jump	0.016*** (0.001)	0.005*** (0.000)	29.8	70.2
Female sample				
RD Jump	0.021*** (0.001)	0.007*** (0.000)	31.6	68.4
Male sample				
RD Jump	0.013*** (0.001)	0.003*** (0.000)	26.4	73.6
p-val (F=M)	0.000	0.000		

* p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses.

Notes: Donut RD at the 2007 cohort cutoff (excluding 2006) with linear trends on either side. Worker and firm FE are estimates using the AKM model reported in equation (3), and the decomposition is done as outlined in equation (4). RD jumps are estimates using equation (1). Shares are computed by dividing Worker FE, Firm FE, or the remainder component by the overall increase in log hourly relative wages. Relative hourly wages are constructed as outlined in Section A.1.

Table 6: Decomposing wage growth effects into firm and non-firm AKM fixed effects

	RD Jump		Share of wage growth (%)	
	$\Delta \text{Log(hwage)}$	$\Delta \text{Firm FE}$	Firm FE	Non-Firm FE
Whole sample				
RD Jump	0.021*** (0.002)	0.001 (0.001)	5.8	94.2
Female sample				
RD Jump	0.026*** (0.002)	0.004*** (0.001)	14.6	85.4
Male sample				
RD Jump	0.016*** (0.002)	-0.001 (0.001)	-6.6	106.6
p-val (F=M)	0.002	0.004		

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

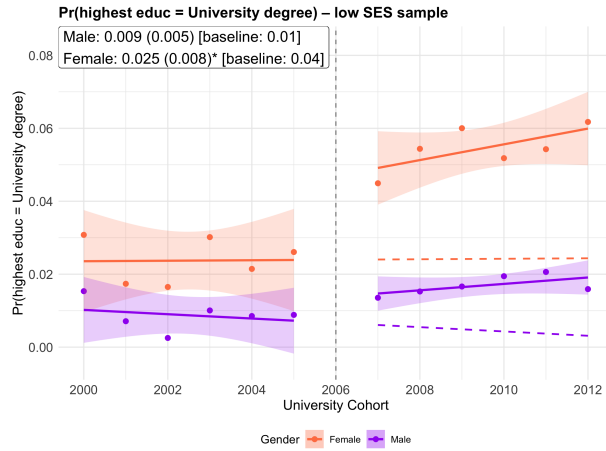
Notes: Donut RD at the 2007 cohort cutoff (excluding 2006) with linear trends on either side. Δ computed as value at entry + 3 years minus value at entry.

Table 7: Decomposing wage growth effects into movers and stayers

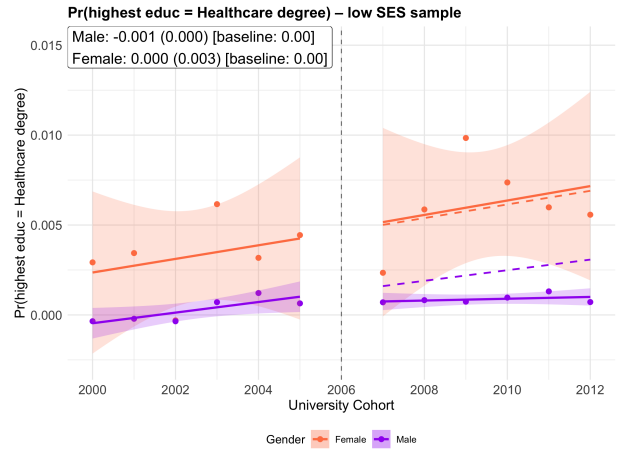
	Pr(changed firm) in first 3 years	$\Delta \text{Log(hwage)} \mid \text{Stayer}$	$\Delta \text{Log(hwage)} \mid \text{Mover}$
Whole sample			
RD Jump	-0.010*** (0.002)	0.021*** (0.002)	0.021*** (0.002)
Female sample			
RD Jump	-0.005 (0.003)	0.028*** (0.003)	0.025*** (0.003)
Male sample			
RD Jump	-0.015*** (0.003)	0.015*** (0.003)	0.018*** (0.003)
p-val (F=M)	0.027	0.000	0.138

* p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses.

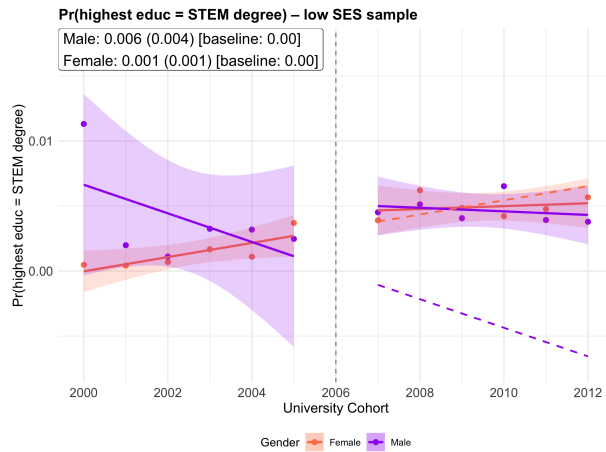
Δ computed as value at entry + 3 years minus value at entry. At baseline, ~50% of sample changes firm between entry and entry +3



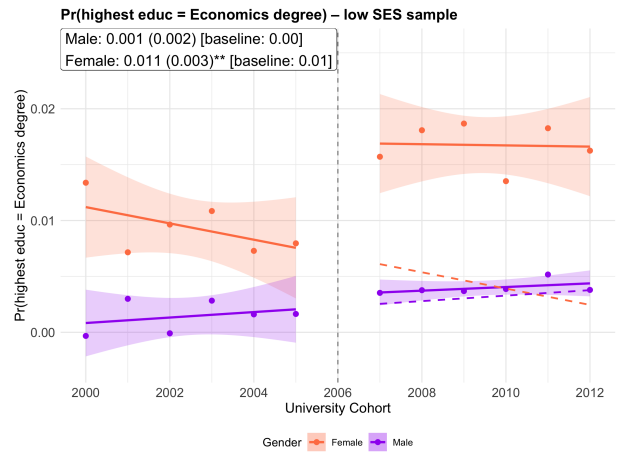
(a) Pr(highest education = university)



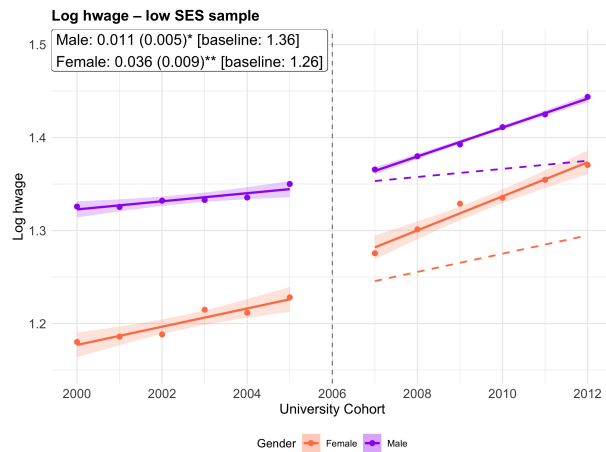
(b) Pr(highest education = university - Healthcare)



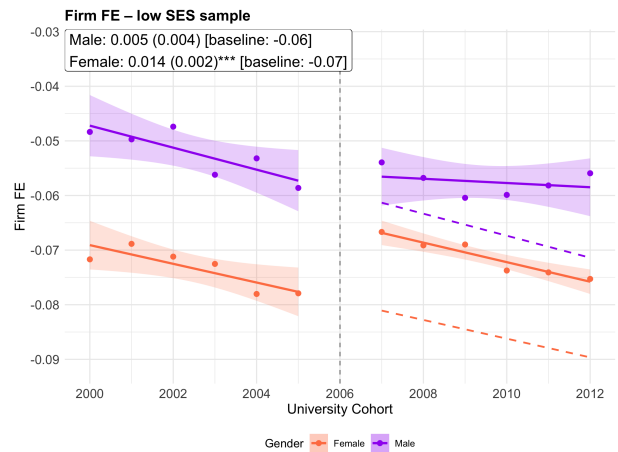
(c) Pr(highest education = university - STEM)



(d) Pr(highest education = university - Economics)



(e) Hourly wage (relative)



(f) Firm AKM fixed effects

Figure 9: RD graphs for individuals who started working before age 18.

B A simple dynamic learning model of university enrolment under the Bologna reform

This section develops a simple dynamic model of university enrolment under uncertainty. The model is designed to capture a central feature of the Bologna reform: the introduction of a valuable intermediate credential after three years of study. The goal is to provide a disciplined framework for thinking about how the reform can simultaneously affect enrolment, dropout, and continuation to full completion.

The key idea is that students are uncertain, at the moment of enrolment, about whether the programme is a good match for them. While enrolled, they receive signals about their programme-specific match quality and update their beliefs. This makes university a sequential learning problem. In the pre-reform regime, students face an all-or-nothing investment: leaving before full completion implies leaving with no university credential. In the post-reform regime, by contrast, students who reach the intermediate threshold can leave with a *Licenciatura* or continue toward full completion. This intermediate exit option lowers the downside risk of trying university and may therefore affect both entry and continuation decisions.

B.1 Environment

Time is discrete and measured in academic years. Individuals enter the model at the moment of high school graduation. At that point, each individual chooses whether to join the labor force immediately or to enrol in university.

The university programme is modelled as a single track. Let

$$s \in \{0, 1, 2, 3, 4, 5\}$$

denote cumulative progress in the programme, where $s = 0$ corresponds to entry and $s = 5$ to full completion. The key difference between the pre- and post-reform regimes concerns the set of options available at the intermediate threshold $s = 3$.

In the **pre-reform regime**, the programme is an all-or-nothing 5-year degree. Students who leave before $s = 5$ obtain no university credential.

In the **post-reform regime**, students who reach $s = 3$ obtain a *Licenciatura*. At that point, they may

either exit to the labor market with the intermediate credential or continue toward full completion at $s = 5$, which corresponds to the Master degree.

B.2 Match quality and beliefs

Students differ in programme-specific match quality. Let

$$\mu \in \{0, 1\}$$

denote the student's true but initially unknown type, where $\mu = 1$ denotes a high-match student and $\mu = 0$ a low-match student.

At the time of high school graduation, the student does not observe μ , but holds an initial prior belief

$$p_0 = \Pr(\mu = 1).$$

More generally, at any point during the programme, the student's belief is denoted by

$$p \in [0, 1].$$

Beliefs evolve through academic signals. In each period, an enrolled student observes a signal

$$\eta \in \{G, B\},$$

where G denotes a good signal and B a bad signal. Conditional on type, the signal probabilities are

$$\Pr(G \mid \mu = 1) = \lambda_H, \quad \Pr(G \mid \mu = 0) = \lambda_L,$$

with

$$\lambda_H > \lambda_L.$$

Thus high-match students are more likely to receive favorable signals.

Given current belief p , the student assigns probability

$$\Pr(G \mid p) = p\lambda_H + (1 - p)\lambda_L$$

to receiving a good signal, and probability

$$\Pr(B \mid p) = 1 - \Pr(G \mid p)$$

to receiving a bad signal.

After observing a signal, the student updates beliefs by Bayes' rule. After a good signal,

$$p'_G = \Pr(\mu = 1 \mid G) = \frac{p\lambda_H}{p\lambda_H + (1-p)\lambda_L},$$

and after a bad signal,

$$p'_B = \Pr(\mu = 1 \mid B) = \frac{p(1-\lambda_H)}{p(1-\lambda_H) + (1-p)(1-\lambda_L)}.$$

B.3 Primitives, state variables, and choices

It is useful to distinguish clearly between *primitives*, *state variables*, and *choices*.

Primitives. The primitives of the model are the objects taken as given by the student:

- the discount factor $\beta \in (0, 1)$;
- the per-period cost of study $\tau > 0$;
- the signal structure (λ_H, λ_L) ;
- the labor-market payoff from leaving without a university credential, denoted w^0 ;
- the return to the Licenciatura, denoted by parameter $\gamma_L > 0$;
- the return to full completion, denoted by parameter $\gamma_M > \gamma_L$.

State variables. The state variables are the objects that summarize the student's current position in the dynamic problem:

- cumulative progress s ;
- current belief p .

Hence the state of an enrolled student is summarized by

$$(s, p).$$

Choices. The student makes the following choices:

- at entry ($s = 0$): whether to work immediately or enrol in university;
- before the intermediate threshold: whether to continue or exit;
- at and after the Licenciatura node in the post-reform regime: whether to exit with the Licenciatura or continue toward full completion.

B.4 Payoffs

The student who enters the labor market without a university credential receives constant per-period payoff

$$w^0.$$

For convenience, the associated lifetime value is

$$W^0 = \frac{w^0}{1 - \beta}.$$

If the student exits with a Licenciatura, the expected per-period payoff depends on current belief p . For simplicity, assume

$$w^L(p) = w^0 + \gamma_L p.$$

The corresponding lifetime value is

$$W^L(p) = \frac{w^L(p)}{1 - \beta} = \frac{w^0 + \gamma_L p}{1 - \beta}.$$

If the student completes the full programme, the expected per-period payoff is

$$w^M(p) = w^0 + \gamma_M p, \quad \gamma_M > \gamma_L,$$

with lifetime value

$$W^M(p) = \frac{w^M(p)}{1 - \beta} = \frac{w^0 + \gamma_M p}{1 - \beta}.$$

Thus, for any given belief p , the value of full completion exceeds the value of exiting with the intermediate credential, which in turn exceeds the value of leaving without a degree:

$$W^M(p) > W^L(p) > W^0.$$

B.5 Timing

Within a period, the timing is as follows:

1. the student enters the period with state (s, p) ;
2. if enrolled, the student observes an academic signal $\eta \in \{G, B\}$;
3. the student updates beliefs from p to p' ;
4. the student chooses whether to continue or exit;
5. if the student continues, progress advances from s to $s + 1$.

In the simplified model developed here, signals affect beliefs but not progress directly. Conditional on continuation, progress evolves deterministically according to

$$s' = s + 1.$$

Thus academic performance matters only because it changes the student's beliefs about match quality, which in turn changes the value of continuing.

B.6 The pre-reform problem

Consider first the pre-reform regime. Let

$$V^{pre}(s, p)$$

denote the value of being enrolled in the programme with progress s and belief p .

Before full completion, the student pays the per-period study cost τ , receives a signal, updates beliefs, and then decides whether to continue or exit with no degree. Since exit yields W^0 , the Bellman equation for all $s < 5$ is

$$V^{pre}(s, p) = -\tau + \beta \mathbb{E} \left[\max \{ V^{pre}(s + 1, p'), W^0 \} \right].$$

The expectation is taken with respect to the signal distribution induced by current belief p . Writing this

explicitly,

$$V^{pre}(s, p) = -\tau + \beta \left[\Pr(G | p) \max\{V^{pre}(s+1, p'_G), W^0\} + \Pr(B | p) \max\{V^{pre}(s+1, p'_B), W^0\} \right], \quad s = 0, 1, 2, 3, 4.$$

At full completion,

$$V^{pre}(5, p) = W^M(p).$$

Hence under the pre-reform regime, the student repeatedly compares the value of continuing with the value of leaving without any university credential.

B.7 The post-reform problem

Now consider the post-reform regime. Let

$$V^{post}(s, p)$$

denote the value of being enrolled in the post-reform programme with state (s, p) .

The Bellman equation differs across three regions: before the Licenciatura threshold, at the Licenciatura threshold, and after the Licenciatura threshold but before full completion.

Before the Licenciatura threshold: $s < 3$. Before reaching $s = 3$, the student still faces the same immediate exit option as in the pre-reform regime: leaving implies no university credential. Hence for $s = 0, 1, 2$,

$$V^{post}(s, p) = -\tau + \beta \mathbb{E} \left[\max \{ V^{post}(s+1, p'), W^0 \} \right].$$

Equivalently,

$$V^{post}(s, p) = -\tau + \beta \left[\Pr(G | p) \max\{V^{post}(s+1, p'_G), W^0\} + \Pr(B | p) \max\{V^{post}(s+1, p'_B), W^0\} \right], \quad s = 0, 1, 2.$$

At the Licenciatura node: $s = 3$. Once the student reaches $s = 3$, the intermediate credential becomes available. At this point the student chooses between:

- exiting immediately with the Licenciatura, which yields $W^L(p)$;

- continuing toward full completion.

The value of continuing is the value of paying the study cost today, then next period facing the continuation problem at $s = 4$, where the fallback option is now the *Licenciatura* rather than no degree. Therefore,

$$V^{post}(3, p) = \max \{ W^L(p), -\tau + \beta \mathbb{E} [\max \{ V^{post}(4, p'), W^L(p') \}] \}.$$

Explicitly,

$$V^{post}(3, p) = \max \left\{ W^L(p), -\tau + \beta [\Pr(G | p) \max \{ V^{post}(4, p'_G), W^L(p'_G) \} + \Pr(B | p) \max \{ V^{post}(4, p'_B), W^L(p'_B) \}] \right\}.$$

After the *Licenciatura* threshold: $3 < s < 5$. Once the student has obtained the *Licenciatura*, the relevant outside option changes permanently. Leaving no longer means leaving with nothing; it means leaving with the intermediate credential. Thus for $s = 4$,

$$V^{post}(4, p) = -\tau + \beta \mathbb{E} [\max \{ V^{post}(5, p'), W^L(p') \}].$$

Equivalently,

$$V^{post}(4, p) = -\tau + \beta \left[\Pr(G | p) \max \{ V^{post}(5, p'_G), W^L(p'_G) \} + \Pr(B | p) \max \{ V^{post}(5, p'_B), W^L(p'_B) \} \right].$$

At full completion,

$$V^{post}(5, p) = W^M(p).$$

B.8 Initial enrolment decision

At the moment of high school graduation, the student compares the value of working immediately to the value of entering university.

Under the pre-reform regime, the initial value is

$$\Lambda^{pre}(p_0) = \max \{ W^0, V^{pre}(0, p_0) \}.$$

Under the post-reform regime, it is

$$\Lambda^{post}(p_0) = \max \{ W^0, V^{post}(0, p_0) \}.$$

The student enrolls if and only if the value of entering university exceeds the outside option:

$$V^{pre}(0, p_0) \geq W^0 \quad \text{or} \quad V^{post}(0, p_0) \geq W^0.$$

This defines an enrolment threshold in beliefs. Since the post-reform regime improves the continuation value at and after the intermediate node, the value of entering university is weakly higher under the post-reform regime. Hence the enrolment threshold falls.

B.9 How the model works

The logic of the model is straightforward.

Under the pre-reform regime, university is an all-or-nothing investment. Students who enrol know that if they later learn the programme is a bad match, their only exit option is to leave without a degree. This makes entry riskier.

Under the post-reform regime, the intermediate credential changes the problem in two ways. First, it lowers the downside risk of entry, because students know they may leave after three years with a valuable credential rather than nothing. Second, it changes continuation incentives after the intermediate threshold, because the relevant outside option becomes the *Licenciatura* rather than no degree.

Beliefs matter because they summarize the student's expected return to continuing. A higher belief p implies a higher expected payoff from both the *Licenciatura* and full completion. Signals matter because they change beliefs, which in turn change the value of continuing relative to exiting.

B.10 Implications

The simplified model yields a number of intuitive implications.

Enrollment. Since the post-reform regime improves the intermediate fallback option, it raises the ex ante value of entering university. Therefore the belief threshold for entry falls, implying higher enrolment.

Dropout before the intermediate threshold. Before $s = 3$, the student's current exit option remains no degree in both regimes. However, in the post-reform regime, reaching $s = 3$ is more valuable because it unlocks the *Licenciatura*. This raises the continuation value before $s = 3$, which may reduce early dropout.

Three-year completion. The post-reform regime creates a new set of students for whom exiting with the *Licenciatura* is preferable to both dropping out earlier and continuing all the way to full completion. Thus a three-year completion margin emerges.

Continuation to full completion. The effect on continuation beyond the intermediate threshold is ambiguous. On the one hand, the *Licenciatura* raises the value of stopping at $s = 3$, which mechanically reduces continuation. On the other hand, the reform draws in new entrants who would not have enrolled before, and some of them may learn through favorable signals that they are good matches and continue all the way to full completion.

Average years conditional on completion. The model does not imply that average years of study among degree holders must fall sharply after the reform. Although some students now stop at the intermediate credential, some newly induced entrants continue to full completion. Thus the overall change in years conditional on completion can be small.

B.11 Interpretation

The model is intentionally simple. It abstracts from assets, borrowing constraints, and endogenous study effort in order to isolate the mechanism of interest: the interaction between uncertainty, learning, and the intermediate exit option created by the Bologna reform. In this sense, the model should be understood not as a full structural description of university behavior, but as a parsimonious framework for interpreting the empirical margins of adjustment induced by the reform.

Its central message is that the Bologna reform can be understood as lowering the downside risk of university enrolment by turning a long all-or-nothing educational investment into a sequential learning problem with a valuable intermediate credential.

B.12 Initial parameterization

To illustrate the mechanism of the model, I begin with a simple baseline parameterization. At this stage, the goal is not to estimate the model structurally, but rather to show that a plausible set of parameter values

can generate the main qualitative patterns of interest. The parameterization should therefore be understood as an initial benchmark that will later be refined and disciplined more tightly using empirical moments.

Time and progress. One period corresponds to one academic year. Progress in the programme is measured by

$$s \in \{0, 1, 2, 3, 4, 5\},$$

where $s = 3$ is the Licenciatura threshold in the post-reform regime and $s = 5$ is full completion.

Preferences and study cost. The annual discount factor is set to

$$\beta = 0.95,$$

and the per-period cost of staying enrolled is

$$\tau = 0.15.$$

The outside-option payoff without a university credential is normalized to

$$w^0 = 1.$$

Signal process. Signals are binary, $\eta \in \{G, B\}$, and the benchmark signal structure is

$$\Pr(G \mid \mu = 1) = \lambda_H = 0.8, \quad \Pr(G \mid \mu = 0) = \lambda_L = 0.3.$$

Thus, high-match students are substantially more likely to receive favorable signals than low-match students, but the signal is still noisy.

Credential payoffs. The benchmark return to the Licenciatura is set to

$$\gamma_L = 0.2,$$

while the return to full completion is set to

$$\gamma_M = 0.5.$$

The associated lifetime values are therefore

$$W^0 = \frac{1}{1 - \beta}, \quad W^L(p) = \frac{1 + \gamma_L p}{1 - \beta}, \quad W^M(p) = \frac{1 + \gamma_M p}{1 - \beta}.$$

This implies that the *Licenciatura* has positive value relative to leaving without a degree, but that full completion remains substantially more valuable.

Initial beliefs. In the numerical illustration, initial beliefs p_0 are placed on a fine grid over $(0, 1)$. For simulation exercises, I begin with a simple benchmark distribution that places nontrivial mass across the entire support, so that the model contains both students who are almost sure they are poor matches and students who are almost sure they are good matches, as well as students with substantial uncertainty. This ensures that the reform can operate through both the entry margin and the continuation margin.

Interpretation of the benchmark. This initial parameterization is chosen to make the key forces of the model transparent. The value of the *Licenciatura* is high enough to lower the downside risk of entering university, but not so high that all students stop at the intermediate credential. Likewise, the signal process is informative enough to generate learning and sorting over time, but not so informative that a student's match quality is revealed immediately after a single year.

Role of the benchmark calibration. Under this parameterization, the model can be used to compute:

- the enrolment threshold before and after the reform;
- the decision regions over (s, p) -space, such as exit, continuation, and intermediate completion;
- simulated dropout, three-year completion, and five-year completion rates;
- average years of study conditional on obtaining a degree.

The resulting effects should be interpreted as model-implied outcomes under a benchmark parameterization rather than as structurally estimated magnitudes.

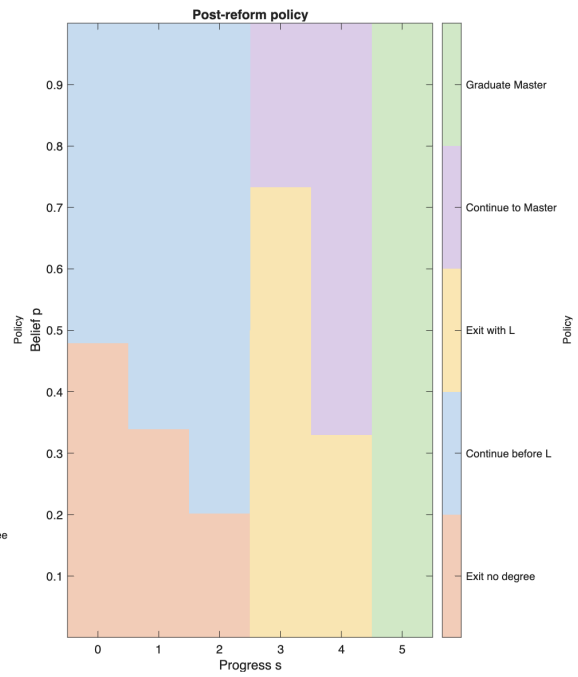
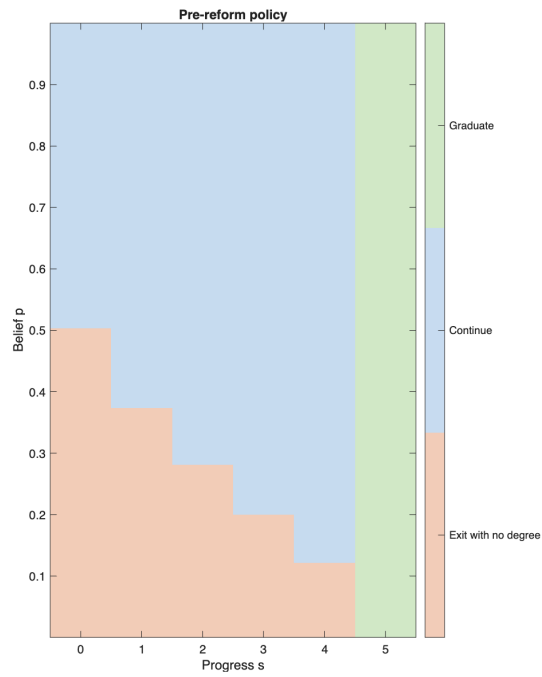
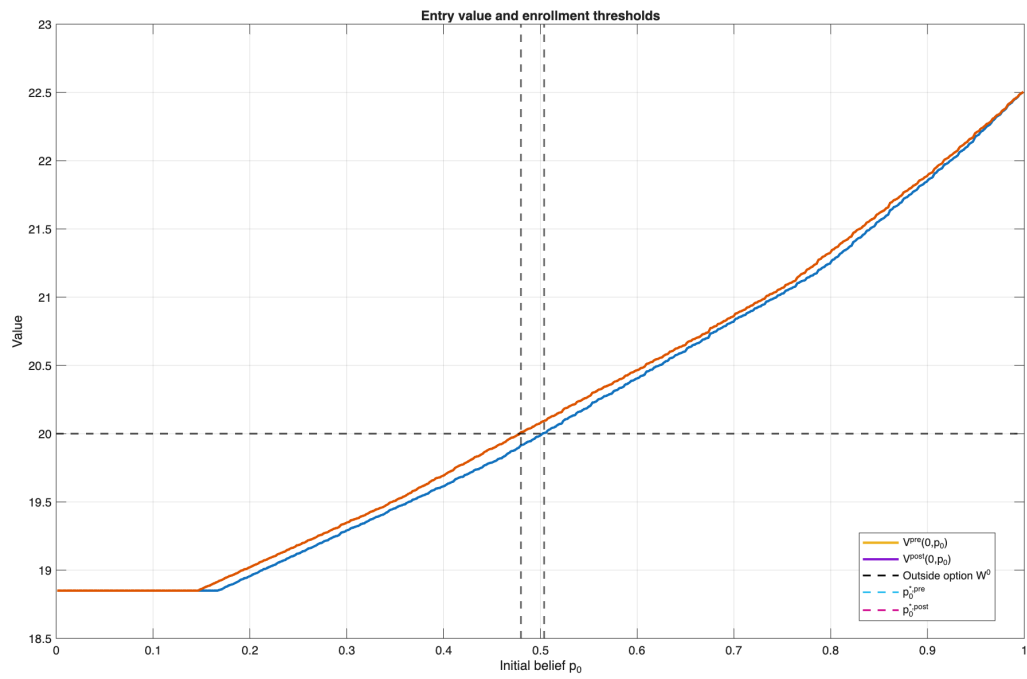
Next step. In future work, the parameterization can be disciplined more formally by selecting parameter values to match key empirical moments, such as the change in enrolment, the change in early dropout, the share of students stopping at three years, the share completing five years, and average years conditional on completion.

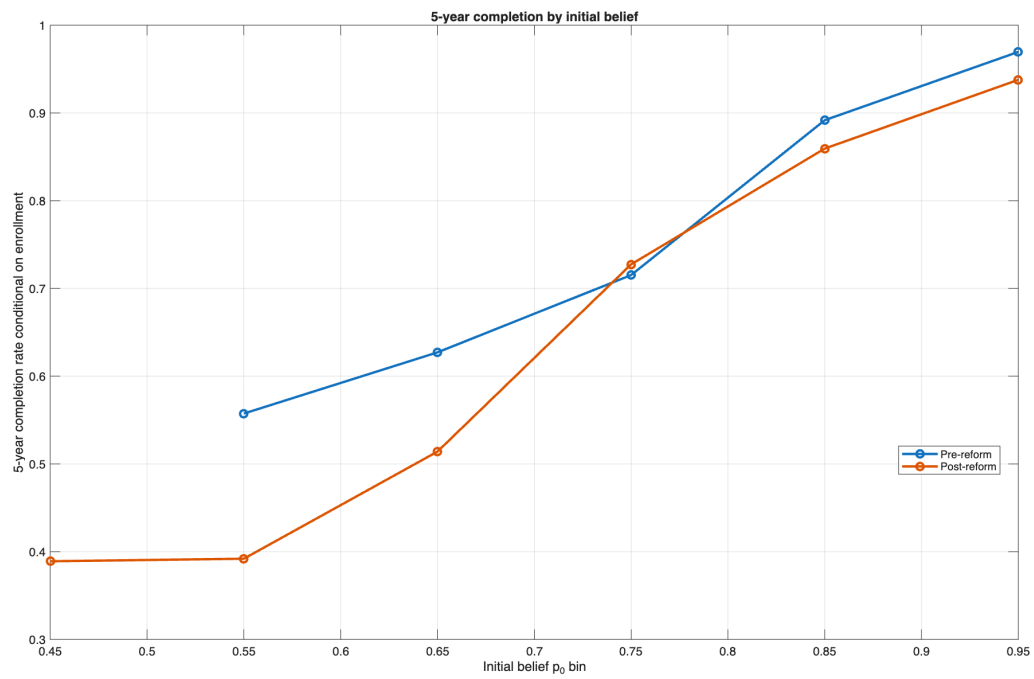
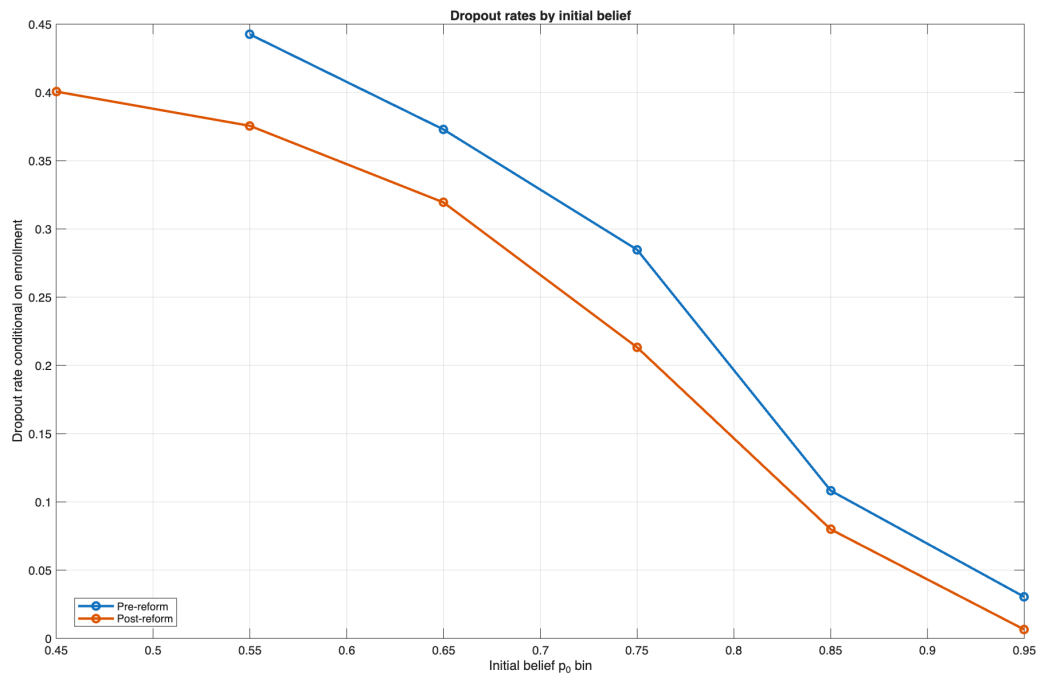
Table 8 summarizes the baseline parameterization used in the initial simulations.

Table 8: Baseline parameterization of the simplified model

Parameter	Value	Interpretation
β	0.95	Annual discount factor
τ	0.15	Per-period study cost
w^0	1.00	Outside-option payoff without degree
λ_H	0.80	$\Pr(G \mid \mu = 1)$
λ_L	0.30	$\Pr(G \mid \mu = 0)$
γ_L	0.20	Return to Licenciatura
γ_M	0.50	Return to full completion

Model initial simulations





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